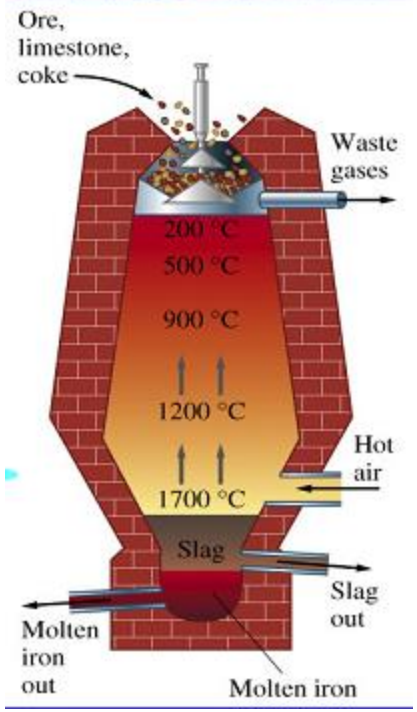
Chapter 19

Chemical Thermodynamics
化学热力学

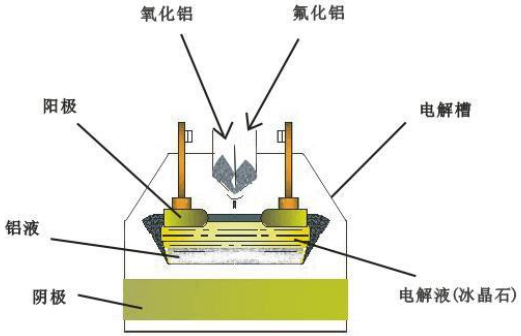
➢ Chemical Thermodynamics (化学热力学)?

$$Fe_2O_3(s) + 3\,CO(g) \rightarrow 2\,Fe(l) + 3\,CO_2(g)$$

为什么不能用同样的方法进行高炉炼铝?



高炉炼铁



- **Chemical kinetics**

  How fast is the reaction?

- **Chemical thermodynamics**

  How far does reaction proceed?

- **化学热力学**

  用热力学原理研究物质体系中的化学现象和规律，根据物质体系的宏观可测性质和热力学函数关系来判断体系的稳定性、变化方向和变化的程度。它是物理化学中较早发展起来的一个学科。

- **化学热力学的作用**

  - 预测反应发生的可能性
  - 判断反应进行的方向(判据)
  - 判断反应进行的限度 (平衡问题)

➢ First Law of Thermodynamics

- You will recall from Chapter 5 that energy cannot be created or destroyed.
- Therefore, the total energy of the **universe** is a constant.
- Energy can, however, be converted from one form to another or transferred from a **system** to the **surroundings** or vice versa.

➢ 化学热力学常用术语

□ 体系和环境

● 体系 (System)
指我们研究的对象，它包含一定种类和一定数量的物质。

● 环境 (Surrounding)
在体系之外又与体系密切相关的部分，称为环境。

- ✓ The **system** is the sample or reaction mixture in which we are interested.
- ✓ Outside the system are the **surroundings**.
- ✓ The system plus its surroundings is sometimes called the **universe**.

□ 封闭体系(closed system)

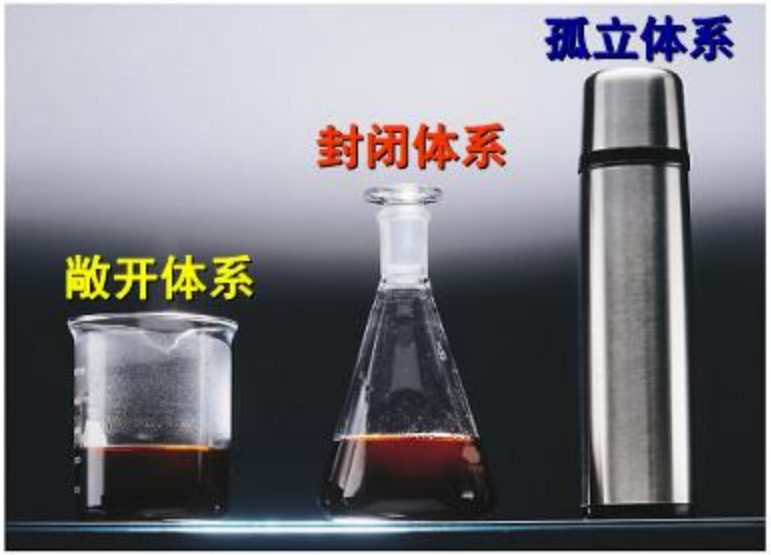
体系和环境之间可以有能量交换而没有物质交换的体系。

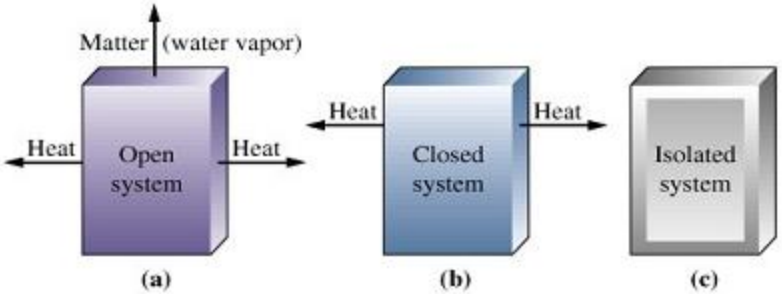
□ 敞开体系 (open system)

体系和环境之间既有能量交换也有物质交换的体系。

□ 孤立体系 (isolated system)

和环境既没有物质交换也没有能量交换的体系。





➢ 热和功

□ 热 (heat，Q)

由于温差导致体系与环境交换能量的一种形式。体系的吸热或放热会导致体系内能的改变。

Q 的符号 ——体系吸热取正值，放热取负值

□ 功 (work, W)：

体系对抗外力运动会作功。提起重物时克服重力作功，气缸中的气体推动活塞作功，电池产生的电流会通过电机作功。

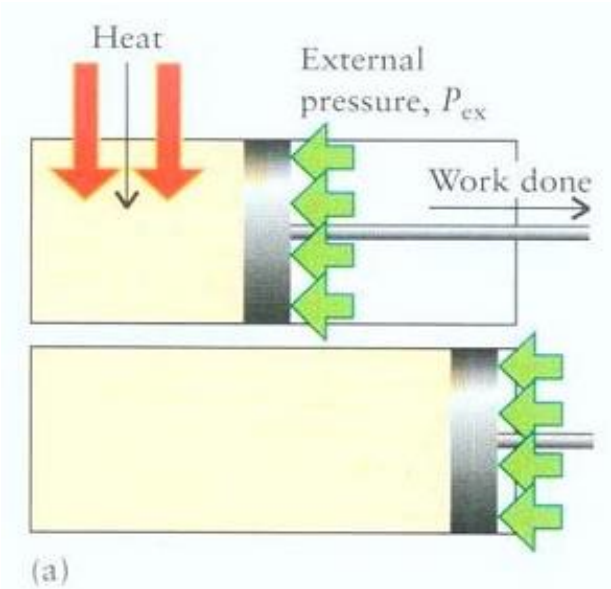
$$\text{Work} = (\text{Force}) \times (\text{Distance})$$

功可以分为机械功、电功、体积膨胀功等各种形式的功。

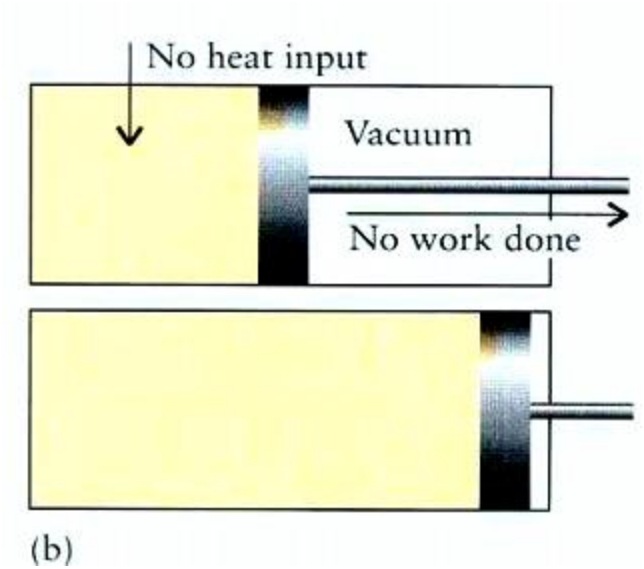
W的符号 —— 环境对体系做功取正值，体系对环境做功取负值

➢ 体积膨胀功

许多常见的化学变化或物理变化往往只做等压体积膨胀功(W_e)而不做其他类型的功 (W')。

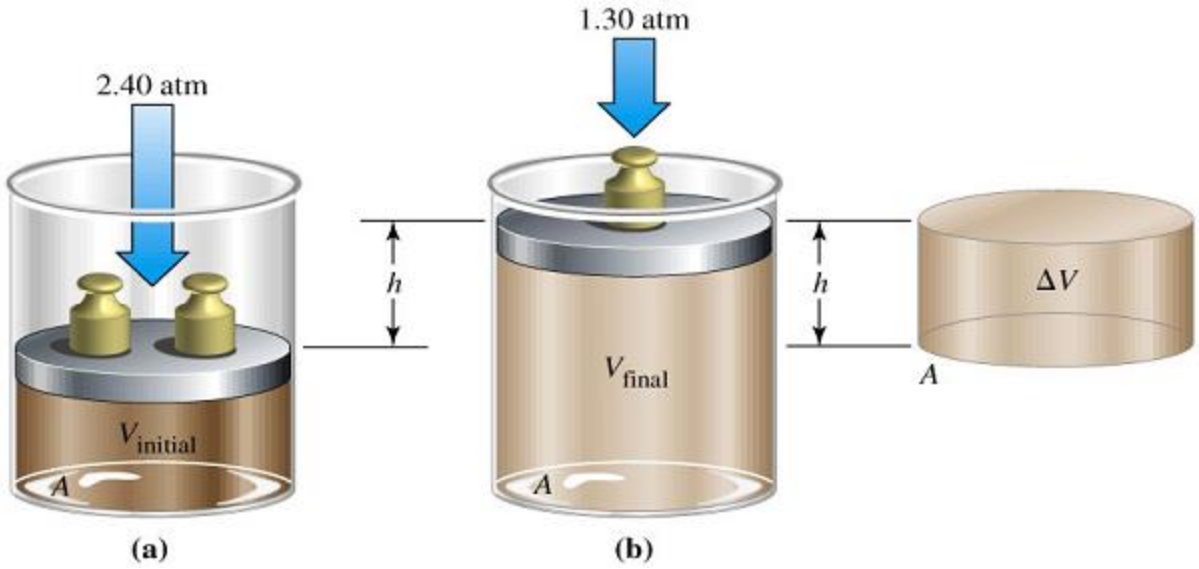


The gas does work as it expands isothermally, but heat flows in to restore the energy lost as work.



The gas does no work as it expands isothermally into a vacuum.

➢ 体积膨胀功的计算



$$W = -F \cdot d = -(P \cdot A) \cdot h$$

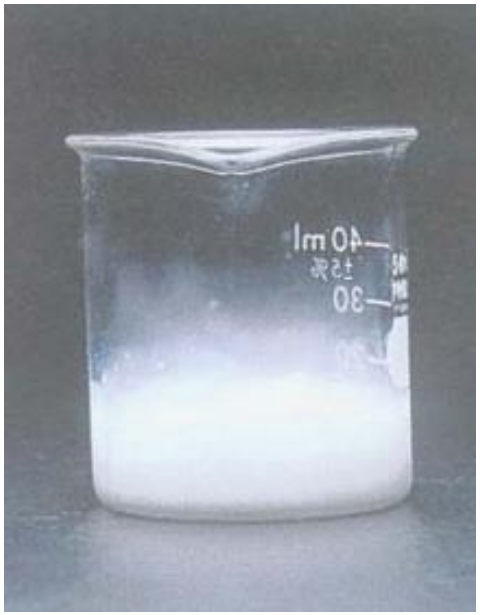
$$= -P \cdot \Delta V$$

$$W = -P_{ext} \cdot \Delta V$$

➢ 反应热的测量

恒压反应热(Q_p)和恒容反应热(Q_v)

铝热剂(thermite)可引发强烈的放热反应($Al + Fe_2O_3$)，其可熔化所产生的金属铁，并产生“铁花”。



NH_4SCN与$Ba(OH)_2 \cdot 8H_2O$的反应强烈吸热，导致反应烧杯外壁能将空气中的水蒸气凝结成冰。

➢ 热力学第一定律与焓的概念

□ 热力学第一定律能量守恒定律的化学表述：

体系内能的变化Δ*E*等于体系从环境所吸收的热量(*Q*)*加上环境对体系所做的功*(*W*)。

$$\Delta E = W + Q$$

式中***W***可以是机械功、电功、体积膨胀功等各种形式的功。

注意：

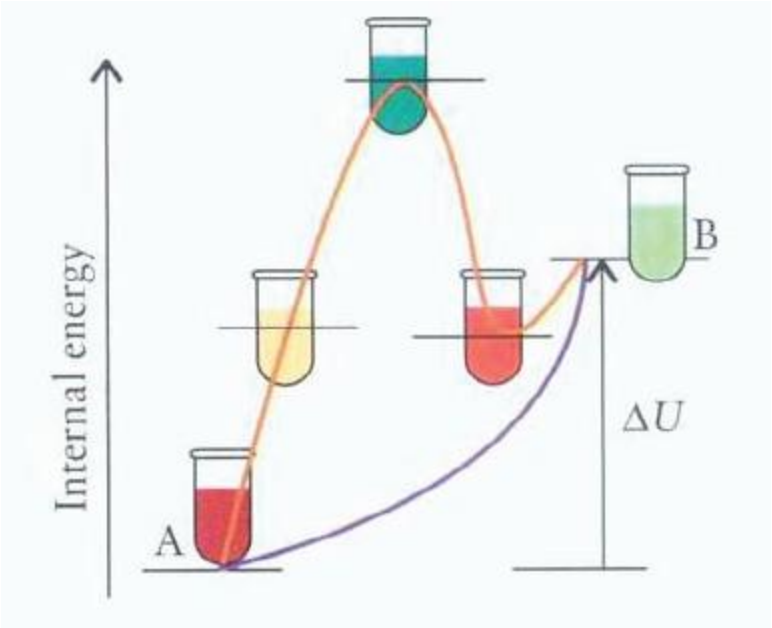
吸热***Q***为正，放热***Q***为负；

环境对体系做功***W***为正，反之***W***为负；

Δ ***E***为正表示体系内能增加，为负表示体系内能减少。

➢ 状态函数(state function):

- 内能这种决定物质状态的物理量称为状态函数。凡只与体系所处状态有关，而与变化路径无关的物理量都是状态函数。例如***p, V, T, U* 等都是状态函数。**
- 各种状态函数之间相互联系相互制约。一定量物质在一定温度和压力下，其体积、内能、焓都是确定的。(数学上的独立变量概念)。
- 热和功都不是状态函数，它们均与变化的路径有关。



Internal energy is a state function: if a system changes from state A to state B (as depicted highly diagrammatically here), the net change in internal energy is the same whatever the route the sequence of chemical or physical changes between the two states.

➢ 焓的物理意义：

- 是人们在处理体系状态变化时引入的一个状态函数，其定义为$H = E + pV$。
- 在等压条件下，体系的焓变等于体系的内能变化和体系所做的等压体积膨胀功之和，即$\Delta H = \Delta E + p\Delta V$。
- 等压变化过程热效应可以直接测定。在等压条件下只做体积膨胀功的体系的焓变等于 Q_p，*即*$\Delta H=Q_p$。*此即焓变的具体物理意义。*
- 对理想气体而言， $p\Delta V= \Delta nRT$，*所以*$\Delta H= \Delta E+\Delta nRT$。
- 多数化学反应的ΔH*和* ΔE*相差不大。*

➢ 热化学方程式：

注明反应热的化学方程式叫做热化学方程式。
化学热力学标法：

$$H_2(g) + 1/2O_2(g) \rightarrow H_2O(l) \quad \Delta_r H_m^{\theta}(298) = -286\ kJ\cdot mol^{-1}$$

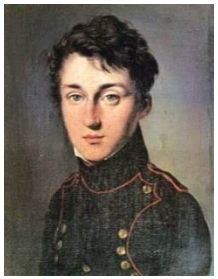
$$1/2N_2(g) + O_2(g) \rightarrow NO_2(g) \quad \Delta_r H_m^{\theta}(298) = +34\ kJ\cdot mol^{-1}$$

其中，左下角的r代表化学反应(reaction)；右下角m代表摩尔(mol)；右上角θ OR o代表热力学标准状态(简称标态)；括号内的数字代表热力学温度，单位为K；Δ*H*代表焓变。

➢ History on Chemical Thermodynamics

Sadi Carnot (1796 – 1832)

Carnot cycle



Rudolph Clausius (1822-1888)

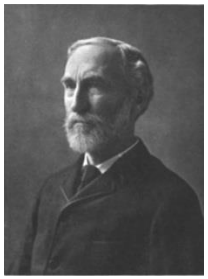
Entropy(熵) $\Delta S = \frac{q}{T}$

Ludwig Boltzmann (1844-1906)

Entropy on the Molecular Level

Josiah Willard Gibbs (1839-1903)

Gibbs free energy



☐ The First Law of Thermodynamics

- ✓ Energy cannot be created or destroyed.
- ✓ Therefore, the **total energy of the universe** is a constant.
- ✓ However, energy can be converted from one form to another or transferred from a system to the surroundings or vice versa.

☐ The Second Law of Thermodynamics

- ✓ The **entropy of the universe** increases in any spontaneous process. ($\Delta S_{univ} > 0$)

☐ The Third Law of Thermodynamics

- ✓ The **entropy of a pure crystalline** substance at absolute zero is 0. $S(0\ K) = 0$

Enthalpy焓/Entropy熵

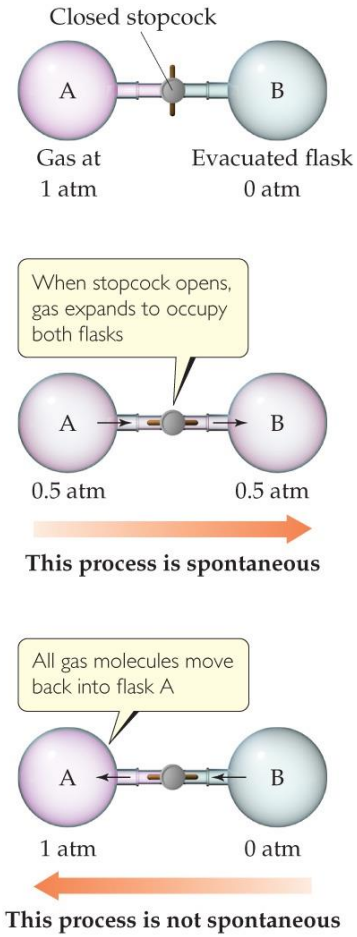
- Change in enthalpy is the heat absorbed by a system during a constant-pressure process.

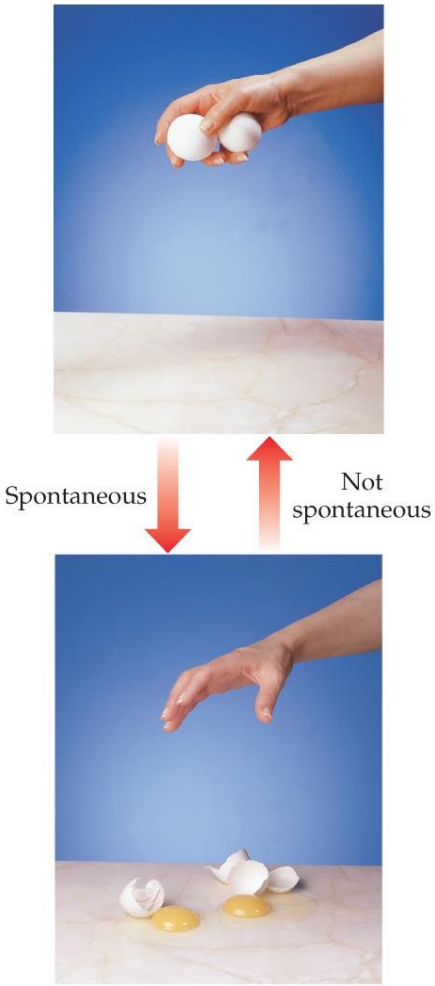
- Entropy is a measure of the randomness in a system.

- Both play a role in determining whether a process is spontaneous自发.

Spontaneous Processes

- **Spontaneous processes** proceed without any outside assistance.
- The gas in vessel A will spontaneously effuse into vessel B, but it will *not* spontaneously return to vessel A.
- Processes that are spontaneous in one direction are nonspontaneous in the reverse direction.

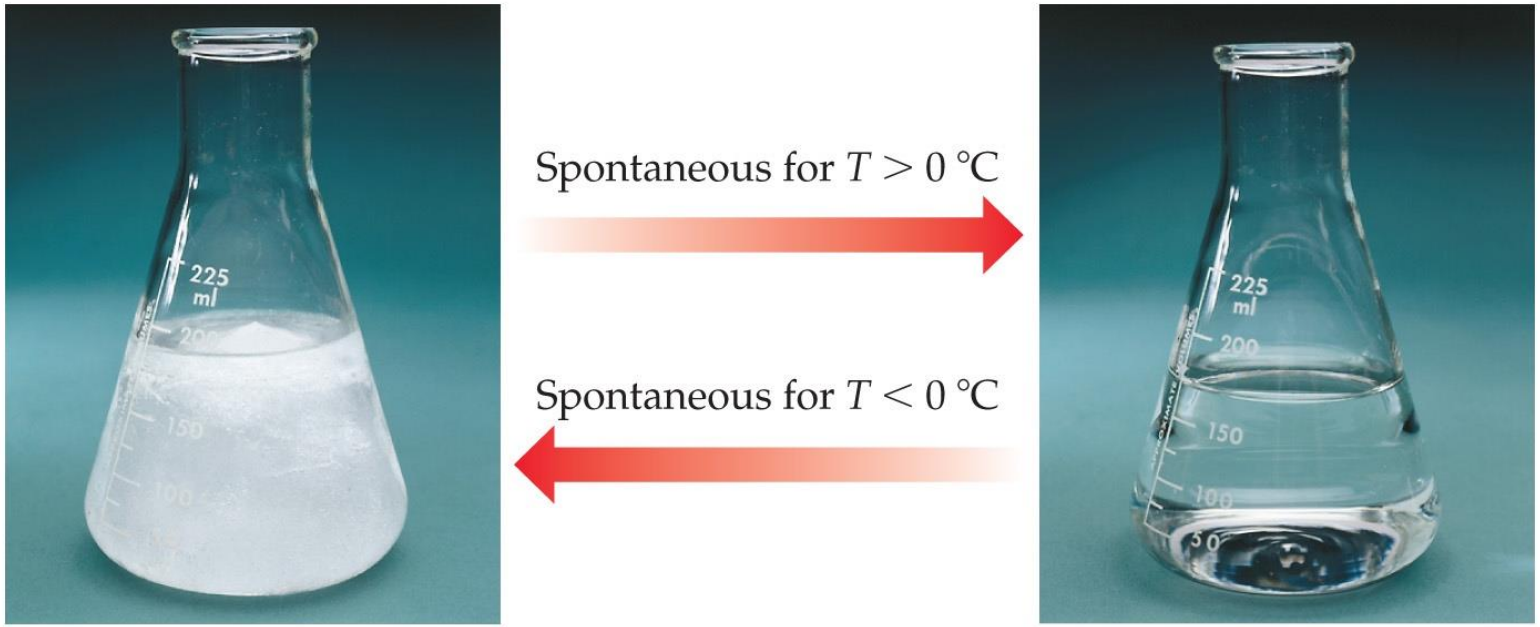




Experimental Factors Affect Spontaneous Processes

- Temperature and pressure can affect spontaneity.

An example of how temperature affects spontaneity is ice melting or freezing.



GIVE IT SOME THOUGHT

If a process is nonspontaneous, does that mean the process cannot occur under any circumstances?

A. Yes. Nonspontaneous processes can never occur under any circumstances.

B. No. Nonspontaneous processes can occur with some continuous external assistance.

➢ Seeking a Criterion for Spontaneity

- Early chemists proposed that spontaneous chemical reactions should occur in the direction of ***decreasing energy***.
- It is true that many ***exothermic*** processes are spontaneous and that many ***endothermic*** reactions are nonspontaneous.
- However, enthalpy change is not a ***sufficient*** criterion for predicting spontaneous change …
  - ✓ The dissolving of NH_4NO_3, are endothermic
  - ✓ The melting of ice at room temperature

Exercise Identifying Spontaneous Processes

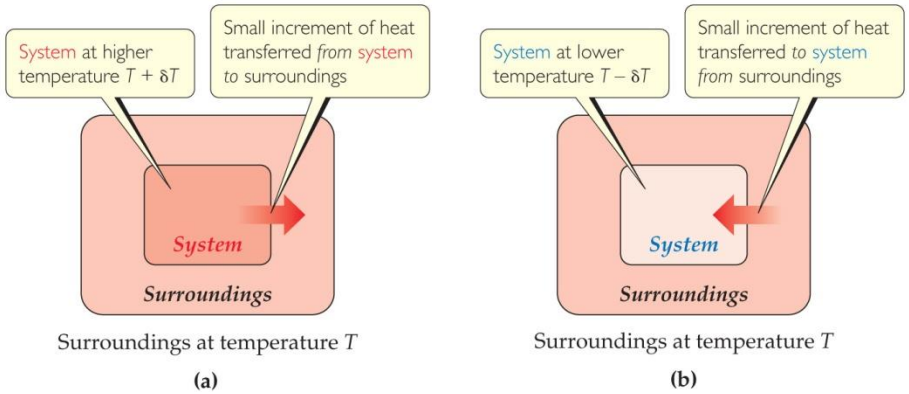
Predict whether each process is spontaneous as described, spontaneous in the reverse direction, or in equilibrium: **(a)** Water at 40 °C gets hotter when a piece of metal heated to 150 °C is added. **(b)** Water at room temperature decomposes into $H_2(g)$ and $O_2(g)$. **(c)** Benzene vapor, $C_6H_6(g)$, at a pressure of 1 atm condenses to liquid benzene at the normal boiling point of benzene, 80.1 °C.

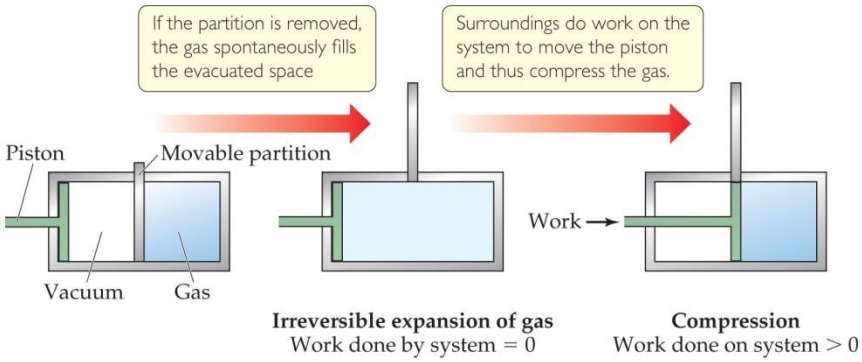
Solution

Solve

(a) This process is spontaneous. Whenever two objects at different temperatures are brought into contact, heat is transferred from the hotter object to the colder one. (Section 5.1) Thus, heat is transferred from the hot metal to the cooler water. The final temperature, after the metal and water achieve the same temperature (thermal equilibrium), will be somewhere between the initial temperatures of the metal and the water. **(b)** Experience tells us that this process is not spontaneous—we certainly have never seen hydrogen and oxygen gases spontaneously bubbling up out of water! Rather, the *reverse* process—the reaction of H_2 and O_2 to form H_2O—is spontaneous. **(c)** The normal boiling point is the temperature at which a vapor at 1 atm is in equilibrium with its liquid. Thus, this is an equilibrium situation. If the temperature were below 80.1 °C, condensation would be spontaneous.

Reversible and Irreversible Processes





Reversible process: The system changes so that the system and surroundings can be returned to the original state by exactly reversing the process. This maximizes work done by a system *on* the surroundings.

Irreversible processes cannot be undone by exactly reversing the change to the system or cannot have the process exactly followed in reverse. Also, any spontaneous process is irreversible!

➢ Reversible Processes (可逆过程)

- In thermodynamics, a reversible process is a process whose direction can be "reversed" by inducing **infinitesimal changes** to some property of the system via its surroundings.
- Throughout the entire reversible process, the system is in **thermodynamic equilibrium** with its surroundings.
- Having been reversed, it leaves no change in either the system or the surroundings. Since it would take an infinite amount of time for the reversible process to finish, **perfectly reversible processes are impossible**.

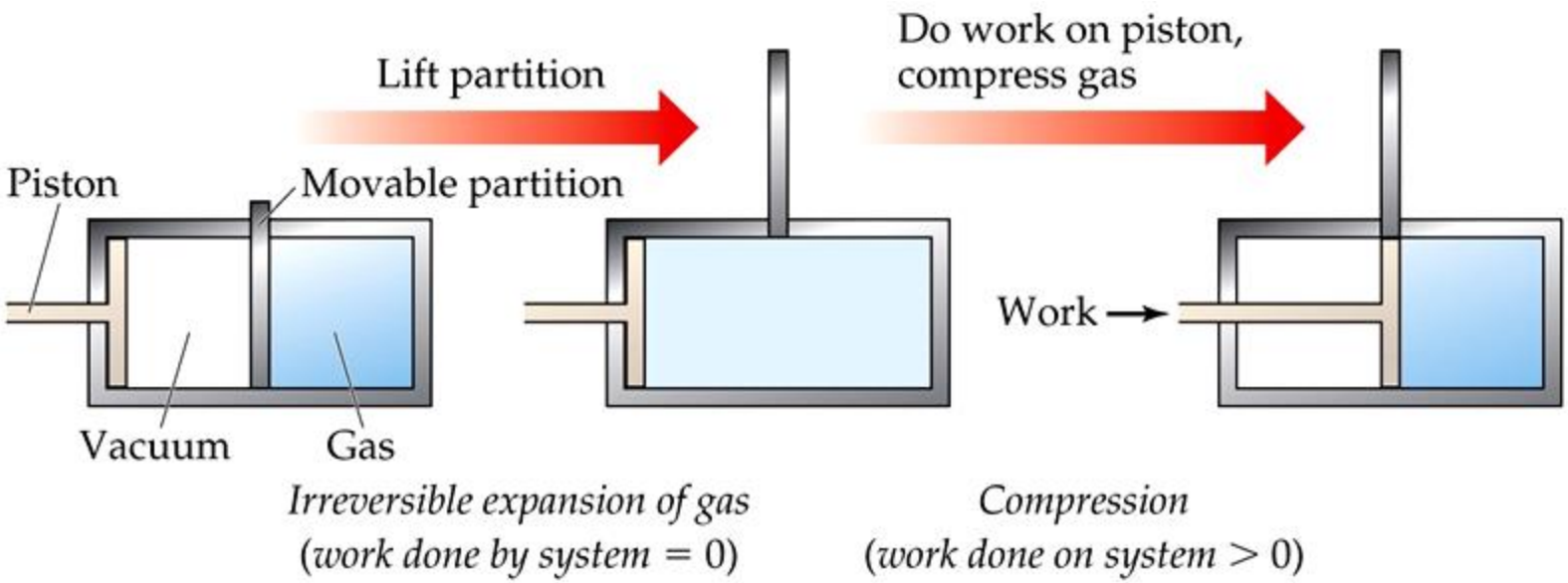
（1）可逆过程是以无限小的变化进行的，整个过程是由一连串非常接近于平衡态的状态所构成。

（2）在反向的过程中，用同样的手续，循着原来的过程的逆过程，可以使系统和环境完全恢复到原来的状态，而无任何耗散效应。

（3）在等温可逆膨胀过程中系统对环境做最大功，在等温可逆压缩过程中环境对系统做最小功。

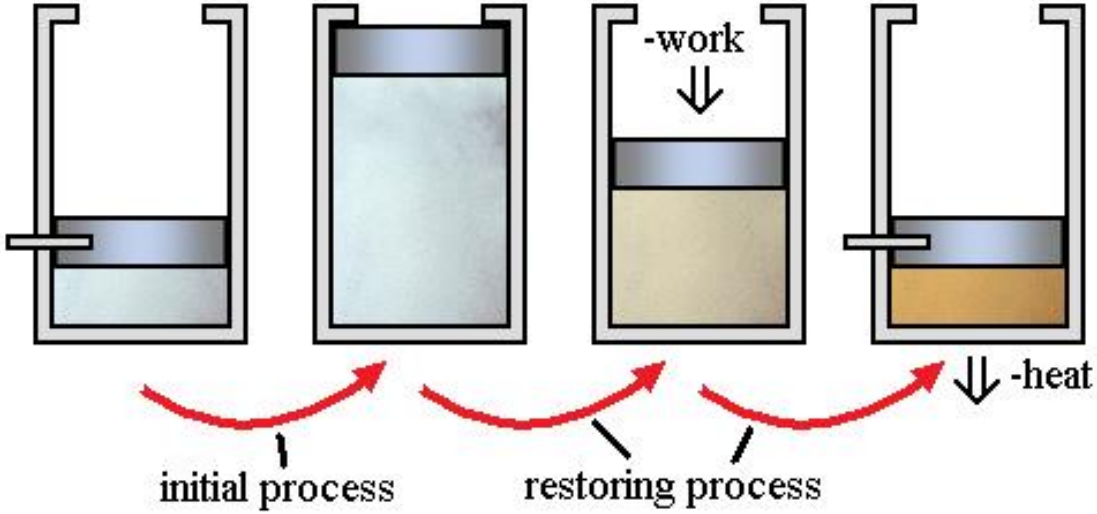
➢ Irreversible Processes (不可逆过程)

- Irreversible processes cannot be undone by exactly reversing the change to the system.
- Spontaneous processes are irreversible.



➢ Irreversible Processes

- An irreversible process is a thermodynamic process that departs from equilibrium. In terms of pressure and volume, it occurs when the pressure (or the volume) of a system changes dramatically and instantaneously that the volume (or the pressure) do not have the time to reach equilibrium.



GIVE IT SOME THOUGHT

Suppose you have a system made up of water only, with the container and everything beyond being the surroundings. Consider a process in which the water is first evaporated and then condensed back into its original container. Is this two-step process necessarily reversible?

A. Yes, because the system is restored to its original state.

B. Yes, because the surroundings are also restored to their original state.

C. No, because there is no evidence that the surroundings are also restored to their original state.

D. No, because energy had to be used to restore the system.

Entropy

- **Entropy** can be thought of as a measure of the randomness of a system.
- It is a state function:

$$\Delta S = S_{\text{final}} - S_{\text{initial}}$$

- It can be found by heat transfer from surroundings at a given temperature:

$$\Delta S = \frac{q_{\text{rev}}}{T} \qquad (\text{constant } T)$$

➢ 熵的物理意义

Carnot

- 熵是体系混乱度的量度。体系越混乱，熵值越大。实际上，熵变也可以从宏观热力学的角度来定义。十九世纪， Carnot 在研究影响热机效率的因素时，提出了Carnot 循环的概念，并提出了著名的卡诺定理(1824年)。

在此基础上，Clausius等人(1854) 总结提出以下关系式：

$$\Delta S = \frac{q_{rev}}{T} \quad \text{Or} \quad q_{rev} = T\Delta S$$

- 上式是熵差的定义式，其中q_{rev}为可逆过程所吸收的热量，体系的熵变等于该**可逆过程**所吸收的热除以温度，熵的中文含义即由其定义“热温商”而得名，而其英文原名的含义是“转变”，指热量可以转变为功的程度。$T\Delta S$ 也可看作是功的一种形式。
- 上式表示在温度T时进行一无限小的可逆过程，吸收微量q_{rev}，使体系的熵增加无限小ΔS，则ΔS等于q_{rev}除以绝对温度 T。

Second Law of Thermodynamics

- The entropy of the universe increases in any spontaneous processes.

- This results in the following relationships:

Reversible Process:	$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} = 0$
Irreversible Process:	$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0$

➢ ΔS for phase change

☐ Consider the melting of ice:

The process is reversible because we can reverse it by infinitely slowly removing the same amount of heat, ΔH_{fusion}, from the system, using immediate surroundings that are infinitesimally below 0° C.

$$q_{rev} = \Delta H_{fusion}$$

The enthalpy of fusion for H_2O is ΔH_{fusion} = 6.01 KJ/mol

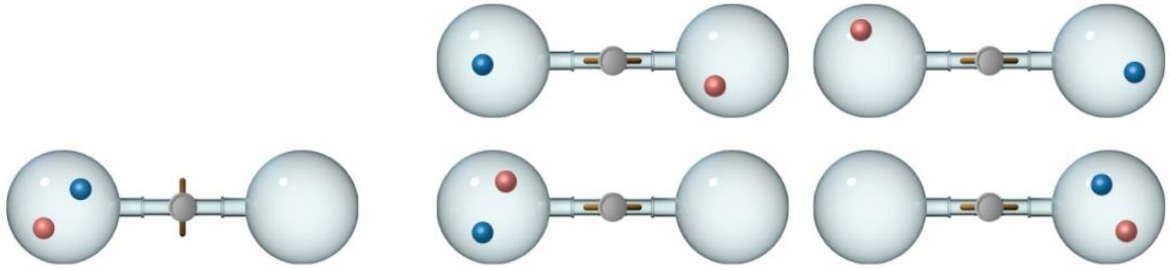
$$\Delta S_{fusion} = \frac{q_{rev}}{T} = \frac{\Delta H_{fusion}}{T} = \frac{(1\ \text{mol})(6.01 \times 10^3\ \text{J/mol})}{273\ \text{K}} = 22.0\ \text{J/K}$$

- 熵是状态函数，这可以从数学上给以证明。根据热力学第一定律，

$$dS = \delta Q_r/T = (dU + pdV)/T$$

- 由于E和V都是体系的广度量，所以S也是体系的广度量，单位是$J \cdot K^{-1}$。
- 须指出，尽管Q不是状态函数，与路径有关，但可逆过程的Qr与路径无关。

Entropy on the Molecular Scale



(a) The two molecules are colored red and blue to keep track of them.

(b) Four possible arrangements (microstates) once the stopcock is opened.

- Boltzmann described entropy on the molecular level.
- Gas molecule expansion: Two molecules are in the apparatus above; both start in one side. What is the likelihood they both will end up there? $(1/2)^2$
- If one mole is used? $(1/2)^{6.02 \times 10^{23}}$! (*No* chance!)
- Gases spontaneously expand to fill the volume given.
- Most probable arrangement of molecules: approximately equal molecules in each side

Statistical Thermodynamics 统计热力学

- Thermodynamics looks at bulk properties of substances (the *big* picture).
- We have seen what happens on the molecular scale.
- How do they relate?
- We use statistics (probability) to relate them. The field is called **statistical thermodynamics**.
- Microstate: A single possible arrangement of position and kinetic energy of molecules

➢ 熵的概念

Ludwig Boltzmann

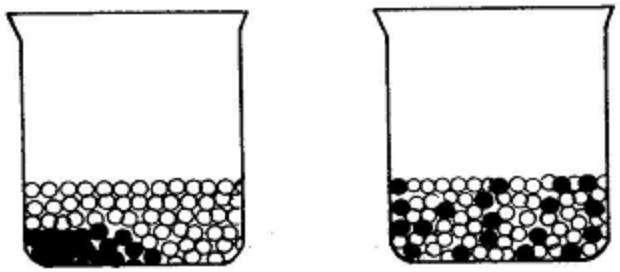
- 为了表述体系的混乱度，人们引入了熵的概念 (*S*, 最早由Clausius提出)。可以把熵看作是体系混乱度(或有序度)的量度，也是一种热力学状态函数。
- 显然，混乱度与体系中可能存在的微观状态数目(w)有关，即有：

$$S = f(w)$$

- 体系中可能存在的微观状态数越多，体系的外在表现就越混乱，熵也就越大。Boltzmann (1877)用统计热力学方法 证明S和*w*呈以下对数关系，即

$$S = k \ln w$$

式中*k*是Boltzmann常数，且$k = R / N_A$。



状态I　　状态II

由状态I变成状态II，混乱度增加

Boltzmann's Use of Microstates

- Because there are so many possible microstates, we can't look at every picture.
- *W* represents the number of microstates.
- Entropy is a measure of how many microstates are associated with a particular macroscopic state.
- The connection between the number of microstates and the entropy of the system is:

$$\boxed{S = k \ln W}$$

Entropy Change

- Since entropy is a state function, the final value minus the initial value will give the overall change.
- In this case, an increase in the number of microstates results in a positive entropy change (more disorder).

$$\Delta S = k \ln W_{\text{final}} - k \ln W_{\text{initial}} = k \ln \frac{W_{\text{final}}}{W_{\text{initial}}}$$

Effect of Volume and Temperature Change on the System

- If we increase volume, there are more positions possible for the molecules. This results in more microstates, so increased entropy.

- If we increase temperature, the average kinetic energy increases. This results in a greater distribution of molecular speeds. Therefore, there are more possible kinetic energy values, resulting in more microstates, increasing entropy.

Molecular Motions 分子运动

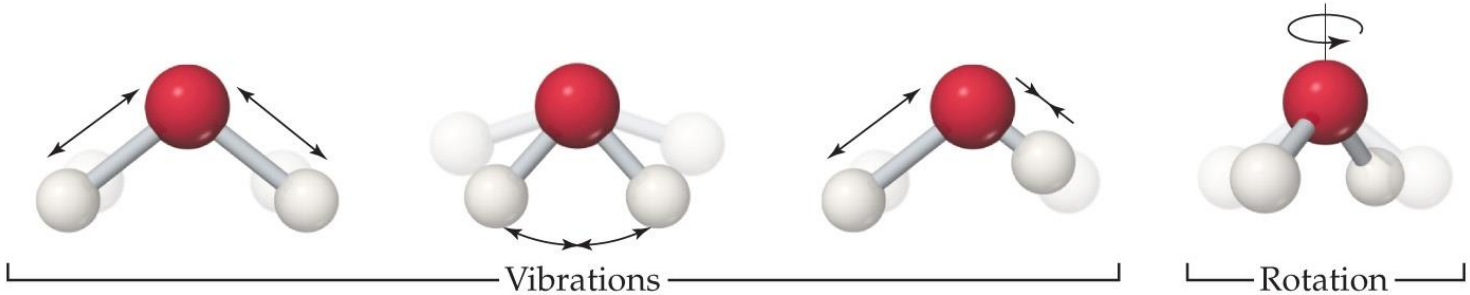
- Molecules exhibit several types of motion.

➢ Translational 平移: Movement of the entire molecule from one place to another

➢ Vibrational 振动: Periodic motion of atoms within a molecule (change in <u>bond lengths or angles</u>)

➢ Rotational 转动: Rotation of the molecule about an axis

- Note: More *atoms* means more microstates (more possible molecular motions).



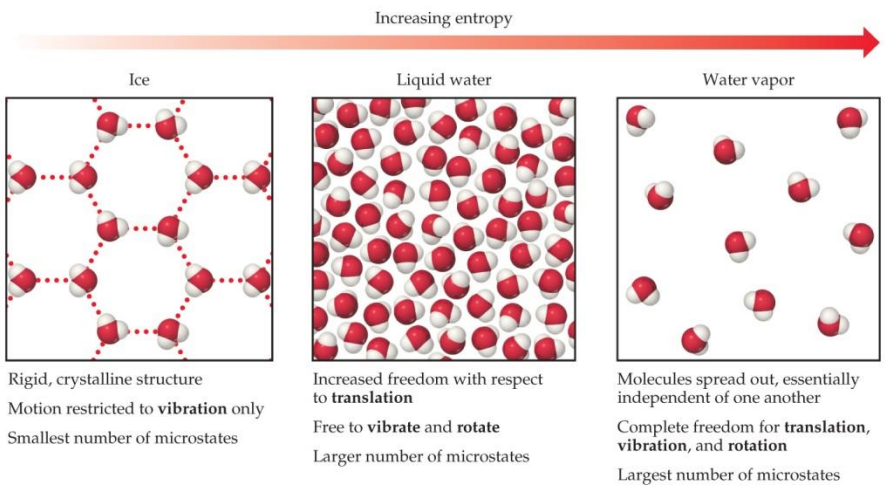
Entropy on the Molecular Scale

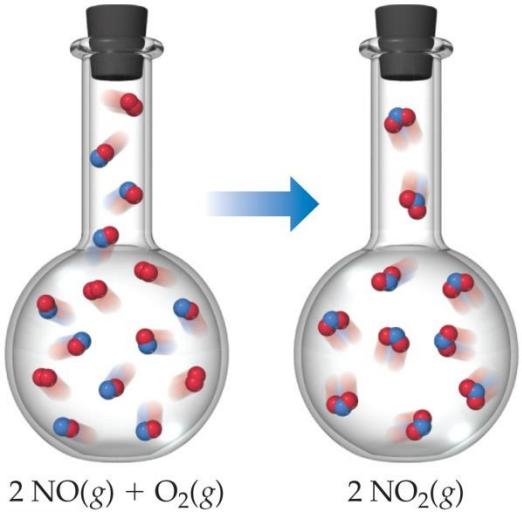
The number of microstates and, therefore, the entropy tend to increase with increases in

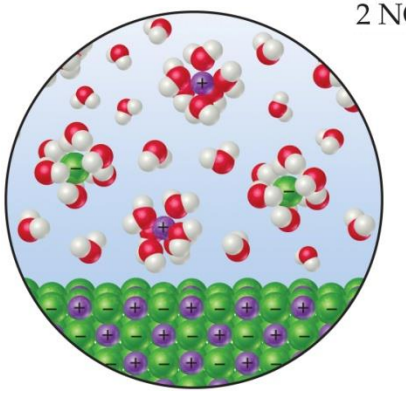
- temperature.
- volume.
- the number of independently moving molecules.

Entropy and Physical States

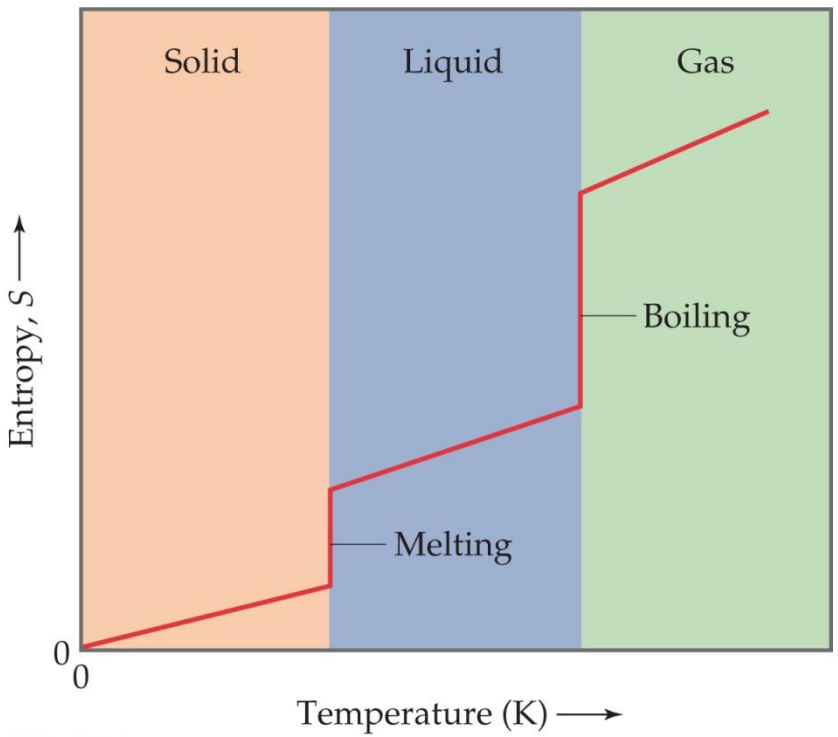
- Entropy increases with the freedom of motion of molecules.
- $S(g) > S(l) > S(s)$
- Entropy of a system increases for processes where
  - ➢ gases form from either solids or liquids.
  - ➢ liquids or solutions form from solids.
  - ➢ the number of gas molecules increases during a chemical reaction.







Third Law of Thermodynamics



- The entropy of a pure crystalline substance at absolute zero is 0.
- Consider all atoms or molecules in the perfect lattice at 0 K; there will only be *one* microstate.
- $S = k \ln W = k \ln 1 = 0$

Standard Entropies 标准熵

standard: per mole

- The reference for entropy is 0 K, so the values for elements are *not* 0 J/mol K at 298 K.
- Standard molar enthalpy for gases are generally greater than liquids and solids. (Be careful of size!)
- Standard entropies increase with molar mass.
- Standard entropies increase with number of atoms in a formula.

Table 19.1 Standard Molar Entropies of Selected Substances at 298 K

Substance	S° (J/mol-K)
$H_2(g)$	130.6
$N_2(g)$	191.5
$O_2(g)$	205.0
$H_2O(g)$	188.8
$NH_3(g)$	192.5
$CH_3OH(g)$	237.6
$C_6H_6(g)$	269.2
$H_2O(l)$	69.9
$CH_3OH(l)$	126.8
$C_6H_6(l)$	172.8
$Li(s)$	29.1
$Na(s)$	51.4
$K(s)$	64.7
$Fe(s)$	27.23
$FeCl_3(s)$	142.3
$NaCl(s)$	72.3

Entropy Changes

Entropy changes for a reaction can be calculated in a manner analogous to that by which ΔH is calculated:

$$\Delta S^\circ = \Sigma nS^\circ \text{ (products)} - \Sigma mS^\circ \text{ (reactants)}$$

S°: standard entropy

where n and m are the coefficients in the balanced chemical equation.

Entropy Changes in Surroundings

- Heat that flows into or out of the system changes the entropy of the surroundings.
- For an isothermal process

$$\Delta S_{\text{surr}} = \frac{-q_{\text{sys}}}{T}$$

- At constant pressure, q_{sys} is simply $\Delta H°$ for the system.

$$\Delta S_{\text{surr}} = \frac{-\Delta H_{\text{sys}}}{T} \quad [\text{at constant P}]$$

Entropy Change in the Universe

- The universe is composed of the system and the surroundings.

- Therefore,

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surroundings}$$

- For spontaneous processes

$$\Delta S_{universe} > 0$$

Total Entropy and Spontaneity 自发性

- $\Delta S_{universe} = \Delta S_{system} + \Delta s_{surroundings}$

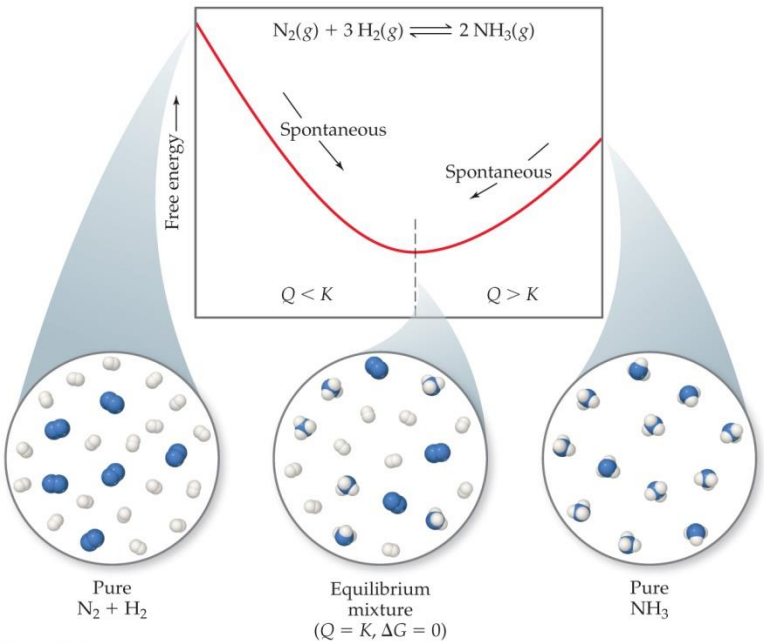
- Substitute for the entropy of the surroundings:
  - $\Delta S_{universe} = \Delta S_{system} - \Delta H_{system}/T$

- Multiply by $-T$:
  - $-T\Delta S_{universe} = -T\Delta S_{system} + \Delta H_{system}$

- Rearrange:
  - $-T\Delta S_{universe} = \Delta H_{system} - T\Delta S_{system} = \Delta(H_{system} - TS_{system})$

- $G = H - TS$, the **Gibbs Free Energy** - a new state function and $\Delta G = \Delta H - T\Delta S$ *(isothermal)*

Gibbs Free Energy



1. If ΔG is negative, the forward reaction is spontaneous ($\Delta s_{universe}$ +).
2. If ΔG is 0, the system is at equilibrium ($\Delta s_{universe}$ 0).
3. If ΔG is positive ($\Delta s_{universe}$ -), the reaction is spontaneous in the reverse direction.

Standard Free Energy Changes

Standard free energies of formation ΔG_f° are analogous (类似) to standard enthalpies of formation (ΔH_f°)

$$\boxed{\Delta G_{rxn}^\circ = \sum n \Delta G_f^o \text{ (products)} - \sum m \Delta G_f^o \text{ (reactants)}}$$

where *n* and *m* are the stoichiometric coefficients.

Free Energy Changes

Table 19.3 How Signs of ΔH and ΔS Affect Reaction Spontaneity

ΔH	ΔS	$-T\Delta S$	$\Delta G = \Delta H - T\Delta S$	Reaction Characteristics	Example
−	+	−	−	Spontaneous at all temperatures	$2\,O_3(g) \longrightarrow 3\,O_2(g)$
+	−	+	+	Nonspontaneous at all temperatures	$3\,O_2(g) \longrightarrow 2\,O_3(g)$
−	−	+	+ or −	Spontaneous at low T; nonspontaneous at high T	$H_2O(l) \longrightarrow H_2O(s)$
+	+	−	+ or −	Spontaneous at high T; nonspontaneous at low T	$H_2O(s) \longrightarrow H_2O(l)$

- How does ΔG change with temperature?
- $\Delta G = \Delta H - T\Delta S$
- Since reactions are spontaneous if $\Delta G < 0$, the sign（符号）of enthalpy and entropy and the magnitude of the temperature matters to spontaneity.

➢ 化学反应自发性的判据：

- 在等温等压的条件下，化学反应的Gibbs自由能变化是焓变和熵变的综合效应，即

$$\Delta G = \Delta H - T\Delta S$$

- ✓ 若 $\Delta H = (-)$，$\Delta S = (+)$时，ΔG必定是负值，即焓降、熵增的化学反应，$\Delta G < 0$，能自发进行。
- ✓ 反之，若 $\Delta H = (+)$，$\Delta S = (-)$ 时，ΔG必定是正值，即焓增、熵降的化学反应，$\Delta G > 0$，不能自发进行。
- ✓ 若 $\Delta G = 0$，体系处于平衡状态。
- ✓ 所以，我们可以用ΔG是正值、还是负值来判别反应的自发性。

GIVE IT SOME THOUGHT

The normal boiling point of benzene is 80 °C. At 100 °C and 1 atm, which term is greater in magnitude for the vaporization of benzene, ΔH or $T\Delta S$?

A. $\Delta H = T\Delta S$

B. $\Delta H < T\Delta S$

C. $\Delta H > T\Delta S$

D. Cannot determine without additional information

Exercise Determining the Effect of Temperature on Spontaneity

The Haber process for the production of ammonia involves the equilibrium

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$$

Assume that ΔH° and ΔS° for this reaction do not change with temperature. (a) Predict the direction in which ΔG° for the reaction changes with increasing temperature. (b) Calculate ΔG° at 25 °C and 500 °C.

Solution

(a) The temperature dependence of ΔG° comes from the entropy term in $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$. We expect ΔS° for this reaction to be negative because the number of molecules of gas is smaller in the products.

Because ΔS° is negative, $-T\Delta S^\circ$ is positive and increases with increasing temperature.

As a result, ΔG° becomes less negative (or more positive) with increasing temperature. Thus, the driving force for the production of NH_3 becomes smaller with increasing temperature.

Exercise Determining the Effect of Temperature on Spontaneity

The Haber process for the production of ammonia involves the equilibrium

$$N_2(g) + 3\,H_2(g) \rightleftharpoons 2\,NH_3(g)$$

Assume that ΔH° and ΔS° for this reaction do not change with temperature.
(a) Predict the direction in which ΔG° for the reaction changes with increasing temperature. (b) Calculate ΔG° at 25 °C and 500 °C.

Solution

(b) We calculated ΔH° and ΔS° for this reaction in to be ΔH° = –92.38 kJ and ΔS° = –198.3 J/K. We assume that these values do not change with temperature.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

At T = 25 °C = 298 K, we have

$$\Delta G^\circ = -92.38\text{ kJ} - (298\text{ K})(-198.3\text{ J/K})\left(\frac{1\text{ kJ}}{1000\text{ J}}\right)$$
$$= -92.38\text{ kJ} + 59.1\text{ kJ} = -33.3\text{ kJ}$$

At T = 500 °C = 773 K, we have

$$\Delta G^\circ = -92.38\text{ kJ} - (773\text{ K})\left(-198.3\,\frac{\text{J}}{\text{K}}\right)\left(\frac{1\text{ kJ}}{1000\text{ J}}\right)$$
$$= -92.38\text{ kJ} + 153\text{ kJ} = 61\text{ kJ}$$

Free Energy and Equilibrium

Under any conditions, standard or nonstandard, the free energy change can be found this way:

$$\Delta G = \Delta G^\circ + RT \ln Q$$

(Under standard conditions, concentrations are 1 *M*, so $Q = 1$ and $\ln Q = 0$; the last term drops out.)

➢ 任意条件下的Gibbs自由能变
——van’ t Hoff等温式

- Gibbs自由能变ΔG是化学反应方向的判据。它是与起始反应物和生成物的量相关的，因此在计算反应自由能变时，必须考虑反应物和生成物的起始浓度。起始浓度不同，所得到的ΔG也不同，反应自发进行的方向也就可能不同。
- 某一反应在温度T时，任意状态的ΔG_T和标准状态的$\Delta G_T^{\ominus}$之间的关系用van’t Hoff等温式来描述(也叫作化学反应的等温式)，是利用所谓的 van’t Hoff平衡箱来推导出来的(假定所有物质都可近似地看作理想气体)。

標準狀態(standard state)，簡稱**標狀態**，是一種為了方便計算體系性質的參考態。常見的體系有：純物質、混合物或溶液。雖然 IUPAC 推薦了一套通用的標準狀態，[1] 但原則上，標準狀態可以任意選取。

在化學中，常定義假想的理想溶液濃度為1 M(mol/L)時為標準狀態。在生物化學則常額外使用pH為7,即10^{-7} mol/l H^+離子濃度為標準狀態[2][3]。

IUPAC 推薦指定標準壓力 $p^{\ominus}$ =10^5 Pa。[4] 嚴格地說，標準狀態並沒有指定溫度。舉個例子，氣體的標準狀態被指定為：單位壓力下(通常是1巴)的理想氣體，而沒有指定具體的溫度。但是，現實中的大多數熱力學數據集都是在特定的溫度下測得的，最常見的是：298.15 K(25.00 ° C；77.00 ° F)或較少見的：273.15 K(0.00 ° C；32.00 ° F)。

體系的標準狀態就是體系熱力學性質(焓、熵、Gibbs自由能)的**參考態**。在標準狀態下，元素的標準摩爾生成焓為零，這使得在相當大的範圍內，其他的熱力學性質都變得容易計算，並能夠以製成表格以便查閱。需要注意的是，物質的標準態不一定是真是存在的：舉個例子，可以計算水蒸氣在溫度為 298.15 K、壓力為 10^5 Pa 時的熱力學性質，雖然此時水以液態存在。但這樣做可以使計算出的熱力學性質表保持自洽。

Free Energy and Equilibrium

- At equilibrium, $\boldsymbol{Q = K}$, and $\boldsymbol{\Delta G = 0}$.
- The equation becomes

$$0 = \Delta G^\circ + RT \ln K$$

- Rearranging, this becomes

$$\boldsymbol{\Delta G^\circ = -RT \ln K}$$

or

$$K = e^{-\Delta G^\circ / RT}$$

➢ Free Energy and Equilibrium

- $\Delta G = 0$ is a criterion for equilibrium at any temperature.
- $\Delta G° = 0$ is a criterion for equilibrium at a ***single temperature***, that temperature at which the equilibrium state has all reactants and products in their standard states.
- ΔG and $\Delta G°$ are related through the reaction quotient, Q :

$$\Delta G = \Delta G° + RT \ln Q$$

- When $\Delta G = 0$, then $Q = K_{eq}$, and the equation becomes:

$$\Delta G° = -RT \ln K$$

➢ 化学反应的限度与化学平衡

化学平衡的概念

根据ΔG值可以判断一个化学反应能否自发发生。但是，即便是自发进行的化学反应，也只能进行到一定限度。因为一般化学反应都是可逆地进行的，即正向和反向反应同时进行。当反应进行到一定程度，正向反应速率和逆向反应速率逐渐相等，反应物和生成物的浓度就不再变化，这种表面静止的状态就叫做“平衡状态”。处在平衡状态的物质浓度称为“平衡浓度 ”。

Exercise Relating ΔG to a Phase Change at Equilibrium

(a) Write the chemical equation that defines the normal boiling point of liquid carbon tetrachloride, $CCl_4(l)$.
(b) What is the value of ΔG° for the equilibrium in part (a)?
(c) Use data from Appendix C and Equation 19.12 to estimate the normal boiling point of CCl_4.

Solution

Solve

(a) The normal boiling point is the temperature at which a pure liquid is in equilibrium with its vapor at a pressure of 1 atm:

$$CCl_4(l) \rightleftharpoons CCl_4(g) \quad P = 1\ \text{atm}$$

(b) At equilibrium, $\Delta G = 0$. In any normal boiling–point equilibrium, both liquid and vapor are in their standard state of pure liquid and vapor at 1 atm. Consequently, $Q = 1$, $\ln Q = 0$, and $\Delta G = \Delta G^\circ$ for this process. We conclude that $\Delta G^\circ = 0$ for the equilibrium representing the normal boiling point of any liquid. (We would also find that $\Delta G^\circ = 0$ for the equilibria relevant to normal melting points and normal sublimation points.)

Exercise Relating ΔG to a Phase Change at Equilibrium

(c) Use data from Appendix C and Equation 19.12 to estimate the normal boiling point of CCl_4.

Solution

(c) Combining Equation 19.12 with the result from part (b), we see that the equality at the normal boiling point, T_b, of $CCl_4(l)$ (or any other pure liquid) is

$$\Delta G^\circ = \Delta H^\circ - T_b \Delta S^\circ = 0$$

Solving the equation for T_b, we obtain $T_b = \Delta H^\circ / \Delta S^\circ$

Strictly speaking, we need the values of ΔH° and ΔS° for the $CCl_4(l)$–$CCl_4(g)$ equilibrium at the normal boiling point to do this calculation. However, we can estimate the boiling point by using the values of ΔH° and ΔS° for CCl_4 at 298 K, which we obtain from Appendix C and some calculations.

$$\Delta H^\circ = (1\text{ mol})(-106.7\text{ kJ/mol}) - (1\text{ mol})(-139.3\text{ kJ/mol}) = +32.6\text{ kJ}$$

$$\Delta S^\circ = (1\text{ mol})(309.4\text{ J/mol-K}) - (1\text{ mol})(214.4\text{ J/mol-K}) = +95.0\text{ J/K}$$

As expected, the process is endothermic ($\Delta H > 0$) and produces a gas, thus increasing the entropy ($\Delta S > 0$). We now use these values to estimate T_b for $CCl_4(l)$:

$$T_b = \frac{\Delta H^\circ}{\Delta S^\circ} = \left(\frac{32.6\text{ kJ}}{95.0\text{ J/K}}\right)\left(\frac{1000\text{ J}}{1\text{ kJ}}\right) = 343\text{ K} = 70\text{ °C}$$

Exercise Calculating the Free–Energy Change under Nonstandard Conditions

Calculate at 298 K for a mixture of 1.0 atm N_2, 3.0 atm H_2, and 0.50 atm NH_3 being used in the Haber process:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

Solution

Solve The partial–pressures form of Equation 15.23 gives

$$Q = \frac{P_{NH_3}{}^2}{P_{N_2} P_{H_2}{}^3} = \frac{(0.50)^2}{(1.0)(3.0)^3} = 9.3 \times 10^{-3}$$

We calculated ΔG° = –33.3 kJ for this reaction. We will have to change the units of this quantity in applying Equation 19.19, however. For the units in Equation 19.19 to work out, we will use kJ/mol as our units for ΔG°, where "per mole" means "per mole of the reaction as written." Thus, ΔG° = –33.3 kJ/mol implies per 1 mol of N_2, per 3 mol of H_2, and per 2 mol of NH_3.

We now use Equation 19.19 to calculate ΔG for these nonstandard conditions:

$$\Delta G = \Delta G^\circ + RT \ln Q \qquad [19.19]$$

$$\begin{aligned}\Delta G &= \Delta G^\circ + RT \ln Q \\ &= (-33.3\ \text{kJ/mol}) + (8.314\ \text{J/mol-K})(298\ \text{K})(1\ \text{kJ}/1000\ \text{J}) \ln(9.3 \times 10^{-3}) \\ &= (-33.3\ \text{kJ/mol}) + (-11.6\ \text{kJ/mol}) = -44.9\ \text{kJ/mol}\end{aligned}$$

Exercise Calculating an Equilibrium Constant from ΔG°

The standard free–energy change for the Haber process at was obtained in Sample Exercise 19.9 for the Haber reaction:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \qquad \Delta G^\circ = -33.3 \text{ kJ/mol} = -33{,}300 \text{ J/mol}$$

Use this value of ΔG° to calculate the equilibrium constant for the process at 25 ° C.

Solution

$$\Delta G^\circ = -RT \ln K$$

$$K = e^{-\Delta G^\circ/RT} = e^{-(-33{,}300 \text{ J/mol})/(8.314 \text{ J/mol-K})(298 \text{ K})} = e^{13.4} = 7 \times 10^5$$

Comment This is a large equilibrium constant, which indicates that the product, NH_3, is greatly favored in the equilibrium mixture at 25 ° C. The equilibrium constants for the Haber reaction at temperatures in the range 300 ° C to 600 ° C, given in Table 15.2, are much smaller than the value at 25 ° C. Clearly, a low–temperature equilibrium favors the production of ammonia more than a high temperature one. Nevertheless, the Haber process is carried out at high temperatures because the reaction is extremely slow at room temperature.

➢ Driving Nonspontaneous—Coupled Reactions

- A nonspontaneous reaction may be ***coupled*** with a spontaneous reaction.
- The decomposition of copper(I) oxide is quite nonspontaneous at room temperature:

$$Cu_2O(s) \longleftrightarrow 2\ Cu(s) + \tfrac{1}{2}\ O_2(g) \qquad \Delta G^\circ_{298} = +149.9\ kJ$$

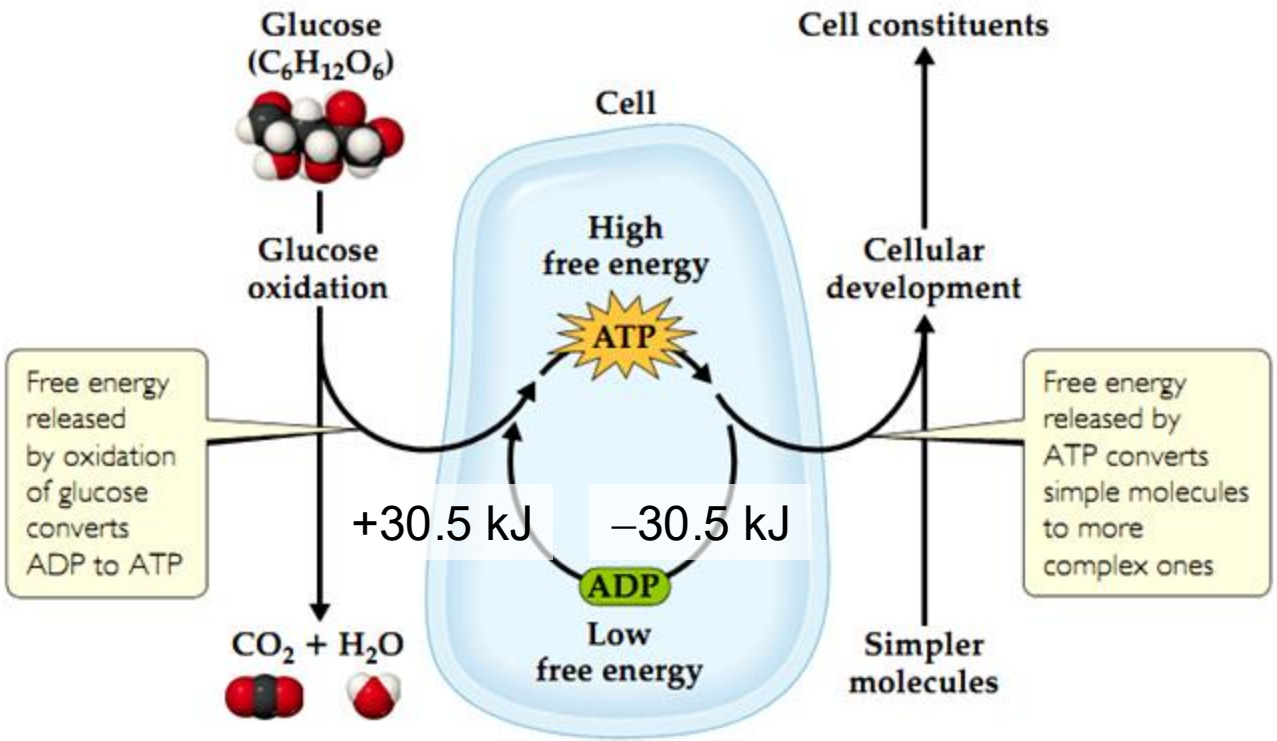
- By coupling this decomposition with the formation of CO from carbon,

$$C(graphite) + \tfrac{1}{2}\ O_2(g) \longleftrightarrow CO(g) \qquad \Delta G^\circ_{298} = -137.2\ kJ$$

we can reduce the nonspontaneity of Cu_2O and make the overall reaction occur slightly above room temperature:

$$Cu_2O(s) + C(graphite) \longleftrightarrow 2\ Cu(s) + CO(g) \qquad \Delta G^\circ_{298} = +12.7\ kJ$$

➢ Coupled Reactions

- Biological systems employ the same principle of using spontaneous reactions to drive nonspontaneous ones.

$$C_6H_{12}O_6(s) + 6O_2(g) \longleftrightarrow 6CO_2(g) + 6H_2O(g) \qquad \Delta G^\circ = -2880 \text{ kJ}$$